ASTRONOMY COMPONENT

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Similar Products

Web site – Solar System Scope

- ➔ https://www.solarsystemscope.com/

You tube shorts

- ➔ https://youtube.com/shorts/ifCa-rzdwMk?feature=shared
- ➔ https://youtu.be/KJynZwHEHnM?feature=shared
- ➔ https://youtube.com/shorts/Eu5ymwk077M?feature=shared
- ➔ https://youtu.be/q9FpWQpw_O8?feature=shared

Research Paper 01

Research Title

- Unlocking the Cosmos: Evaluating the Efficacy of Augmented Reality in Secondary Education Astronomy Instruction

Published Year

- 2024

Research Method

- **Design:** Quasi-experimental with control and experimental groups
- **Approach:** Pretest-posttest design
- **Participants:** 130 secondary students (ages 12–16) in Spain
- **Sampling Method:** Non-probabilistic convenience sampling
- **Data Collection Tool:** Knowledge questionnaire based on Classical Test Theory (CTT)
- **Statistical Methods:** Wilcoxon test, item analysis (difficulty index P, discrimination index D, Cronbach's alpha)

Key Findings

- AR-based instruction significantly improved astronomy learning outcomes compared to traditional methods.
- All four experimental groups (Grades 7–10) showed **statistically significant learning gains**.
- The **control group showed improvement only in one case (Grade 7)**.
- Large **effect sizes** observed in all experimental cases.
- AR was especially effective in helping students visualize abstract astronomical concepts and increased engagement.

Research Gaps Identified

- Most prior AR-in-education studies focus on **lower secondary grades** (like 7th).
- Limited exploration of **AR in complex astronomy topics** beyond the solar system (e.g., light and matter, cosmology).
- Insufficient long-term studies on AR's impact on **motivation and knowledge retention**.
- Lack of **qualitative data** from students' perspectives.
- Most AR content used is **pre-developed**, not tailored with teacher input.

Future Work Recommendations

- Conduct **longitudinal studies** to assess retention and sustained engagement.
- Explore AR's effects on **diverse student demographics** and **learning environments**.
- Investigate **custom-built AR apps** co-designed with teachers.
- Include **mixed-methods research** combining quantitative and qualitative insights.
- Assess AR's integration in other STEM areas beyond astronomy.

Tools and Techniques Used

- **AR Applications:**
  - StarWalk2® / Star Tracker® / Solar System AR Core® / Spot the Station® (NASA) / AR-enabled books: *iSolar System* and *iStorm*
- **Statistical Tools:**
  - Cronbach's alpha / Wilcoxon signed-rank test / Mann–Whitney U test / Effect size (Rank-Biserial Correlation)

Limitations

- Conducted in **only one school in Spain** – generalizability is limited.
- **No long-term** tracking of learning outcomes or motivation.
- Use of **pre-existing AR tools** rather than custom-designed content.
- **No qualitative analysis** of student feedback or experience.
- Did not control for **external variables** like home environment or teacher quality.

Research Paper 02

Research Title

- **Enhancing Astronomy Skills: The Role of Mobile Augmented Reality in Vietnamese Middle Schools**

Published Year

- **2024**

Research Method

- **Design**: Quasi-experimental with control and experimental groups
- **Approach**: Pretest–posttest over 8 weeks
- **Participants**: 438 Grade 6 students (ages 11–12) from 4 schools in Central Vietnam
- **Sampling Method**: Stratified random sampling (balanced by gender, location, and SES)

- **Data Collection Tools**:
    - Astronomy Competency Test (40 MCQs: celestial motion, spatial awareness, conceptual knowledge)
    - Student Perception Questionnaire (SPQ, 20 items)
    - Semi-structured interviews with 40 students
    - **Statistical Methods**: ANCOVA, independent t-tests, Cohen's d (effect size), thematic analysis (qualitative)

Key Findings

- AR group significantly outperformed the control group in celestial motion, spatial awareness, and conceptual knowledge.
- Large effect sizes: Conceptual knowledge showed the strongest effect (Cohen's d = 1.24).
- Female students in the AR group improved more than males in spatial skills, reducing gender disparities.
- Students reported high satisfaction (mean 4.2/5), citing better visualization and engagement.
- Qualitative feedback confirmed AR increased interaction, enjoyment, and conceptual clarity.

Research Gaps Identified

- Limited long-term tracking of knowledge retention.
- Need for more research on infrastructural and socio-cultural barriers in developing countries.
- Few studies linking AR to gender equality outcomes in STEM.
- Lack of teacher-training integration in AR deployment.

Future Work Recommendations

- Conduct longitudinal studies to measure retention and sustained engagement.
- Investigate AR's role in reducing gender gaps in STEM achievement.
- Explore integration strategies in resource-limited settings.
- Co-develop AR with teachers to ensure curriculum alignment.
- Expand AR studies across other STEM disciplines in low-resource contexts.

Tools and Techniques Used

- **AR Applications**: Custom mobile AR apps (not commercial), AR markers, 3D models of planets.
- **Statistical Tools**: IBM SPSS 26, ANCOVA, independent t-tests, Cohen's d.
- **Qualitative Tools**: Semi-structured interviews, thematic analysis (Braun & Clarke's 6-step method).

Limitations

- Focused only on Grade 6 students in one region of Vietnam → limited generalizability.
- Intervention lasted only 8 weeks → no long-term evidence.
- Technical issues reported (device performance, connectivity).
- Limited teacher input in content creation.

Research Paper 03

Research Title

- Augmented Reality Mobile Application to Improve the Astronomy Teaching-Learning Process

Published Year

- 2022

Research Method

- **Design**: Experimental design with random selection of groups
- **Approach**: Pretest–posttest comparison
- **Participants**: 60 elementary students (4th and 6th grade, ~ages 9–12) from the NGO IFEJANT program in Peru, divided into control (n=30) and experimental (n=30) groups
- **Sampling Method**: Random selection design
- **Data Collection Tools**: Pretest–posttest indicators (grades, max/min scores, time-on-task)
- **Statistical Methods**: Descriptive analysis, comparison of averages and distributions

Key Findings

- The AR mobile app (*SolarSystemRA*) significantly improved students' understanding of astronomy concepts (planets, solar system).
- Post-test results showed higher average and maximum scores in the experimental group compared to the control group.
- 66.67% of students scored higher in the post-test than in their pre-test.
- Students using AR demonstrated faster problem-solving and higher engagement.
- The interactive 3D animations made astronomy content more attractive and memorable.

Research Gaps Identified

- Study focused only on basic astronomy topics (mainly the solar system).
- Small sample size (60 students) limits generalizability.
- No long-term tracking of retention or motivation.
- Lack of detailed statistical analysis (mostly descriptive).
- Limited exploration of teacher integration into AR development.

Future Work Recommendations

- Expand study with larger and more diverse student populations.
- Conduct longitudinal research on retention and motivation.
- Apply AR to more complex astronomy and STEM topics.
- Include teacher training and co-design in AR content creation.
- Perform deeper statistical testing (e.g., inferential statistics).

Tools and Techniques

- **AR Application**: *SolarSystemRA* developed using Unity + Vuforia SDK (tablet/mobile).
- **AR Features**: Marker-based 3D visualization of planets and solar system.
- **Methodology**: Mobile-D software development methodology (phases: exploration, initialization, production, stabilization, testing).

Limitations

- Limited to one NGO program in Peru → not broadly representative.
- Small sample size (n=60).
- Only short-term effects measured, no longitudinal tracking.
- Focused narrowly on planetary visualization, not broader astronomy concepts.
- Results based mainly on test score comparisons, with no qualitative feedback.

Research Paper 04

Research Title

- Development of Augmented Reality-Assisted Teaching Materials in Science Subjects: Solar System Topic

Published Year

- 2024

Research Method

- **Design**: Research and Development (R&D) using the Borg & Gall model (steps 1–8: potential → usage trials)
- **Approach**: Pre-experimental, one-group pretest–posttest design
- **Participants**: 46 elementary students in Indonesia
    - Small-scale trial: 12 Grade 6 students
    - Large-scale trial: 34 Grade 5 students
- **Sampling Method**: Purposive sampling (students from SD Negeri 3 Bebengan)
- **Data Collection Tools**:
    - Pretest–posttest (30 MCQs)
    - Questionnaires (teacher and student responses)
    - Observations & interviews
    - Expert validation sheets (media, content, language)
- **Statistical Methods**: N-Gain test, descriptive statistics, validation index

Key Findings

- AR-assisted teaching materials (printed books with 3D AR barcodes via Assemblr Edu app) improved science learning outcomes significantly.
- Average scores increased from 40.68 (pretest) → 80.21 (posttest), with **N-gain = 0.7 (moderate effectiveness)**.
- Students became more motivated and engaged in learning the solar system.
- Teacher and student response questionnaires were very positive.
- Expert validation confirmed materials were valid and feasible: Media (94.5%), Content (92.9%), Language (75%).

Research Gaps Identified

- Study focused only on solar system content in one elementary school.
- No control group for stronger comparison.
- Limited sample size (n=46).
- Short intervention period, no long-term tracking of retention.
- No exploration of AR across other science topics beyond astronomy.

Future Work Recommendations

- Apply AR-assisted teaching materials to other science subjects.
- Use larger and more diverse samples.
- Introduce experimental/control group designs for stronger statistical validity.
- Conduct longitudinal studies to assess knowledge retention.
- Explore teacher co-design in AR teaching materials.

Tools and Techniques Used

- **AR Application**: Assemblr Edu app (3D AR barcode scanning).
- **Teaching Materials**: Printed science book with embedded AR codes for solar system visualization.
- **Development Model**: Borg & Gall (adapted R&D steps).
- **Validation Tools**: Expert panels (media, content, language).

Limitations

- Conducted in a single school in Indonesia → generalizability is limited.
- Small participant size (46 students).
- Focused narrowly on the solar system → limited subject coverage.
- Short-term evaluation only; no evidence of long-term retention.
- Did not involve teachers in AR material creation.

COMPARISON

Feature	Study 1 – Spain (2024)	Study 2 – Vietnam (2024)	Study 3 – Peru (2022)	Study 4 – Indonesia (2024)
Title	Unlocking the Cosmos: Evaluating the Efficacy of AR in Secondary Education Astronomy Instruction	Enhancing Astronomy Skills: The Role of Mobile AR in Vietnamese Middle Schools	AR Mobile Application to Improve the Astronomy Teaching-Learning Process (*SolarSystemRA*)	Development of AR-Assisted Teaching Materials in Science Subjects: Solar System Topic
Level / Participants	130 secondary students (ages 12–16), Grades 7–10	438 Grade 6 students (ages 11–12)	60 elementary students (4th & 6th grade)	46 elementary students (Grade 5 & 6)
Country	Spain	Vietnam	Peru	Indonesia
Method & Design	Quasi-experimental, pretest–posttest with control vs. experimental groups	Quasi-experimental, 8-week intervention, stratified random sampling	Experimental, pretest–posttest with control & experimental groups	R&D (Borg & Gall, steps 1–8), pre-experimental one-group pretest–posttest

Feature	Study 1 – Spain (2024)	Study 2 – Vietnam (2024)	Study 3 – Peru (2022)	Study 4 – Indonesia (2024)
Data Tools	Knowledge questionnaire (Classical Test Theory)	Astronomy Competency Test (40 MCQs), Student Perception Questionnaire, Interviews	Pretest–posttest test scores, time-on-task indicators	Pretest–posttest (30 MCQs), Expert validation, Questionnaires, Interviews
Statistical Methods	Wilcoxon test, Mann–Whitney U, Cronbach's α, Effect size (Rank-Biserial)	ANCOVA, t-tests, Cohen's d, Thematic analysis	Mainly descriptive comparisons	N-Gain test, descriptive statistics, Expert validation index
Key Findings	AR significantly improved outcomes vs. traditional methods; all experimental grades improved; large effect sizes; especially strong for visualizing abstract concepts	AR group outperformed control in celestial motion, spatial awareness, conceptual knowledge; females benefited more in spatial skills; strong engagement	AR app improved student test scores and engagement; 67% of students improved from pre to post-test	AR-assisted teaching materials doubled average scores (40.68 → 80.21); N-Gain = 0.7 (moderate effectiveness); very positive teacher & student responses

Feature	Study 1 – Spain (2024)	Study 2 – Vietnam (2024)	Study 3 – Peru (2022)	Study 4 – Indonesia (2024)
Gaps Identified	Few studies on higher-level astronomy (cosmology, light/matter); lack of long-term motivation/retention studies; little qualitative data; AR tools mostly pre-developed	Lack of long-term data; need for gender-focused studies; little research in developing country contexts; few teacher-integration approaches	Small sample, narrow focus on solar system; no qualitative insights; only short-term results	Small sample, no control group; limited to solar system; short-term only; no teacher involvement in development
Future Work	Longitudinal retention studies; co-design AR with teachers; explore complex astronomy topics; integrate across STEM	Expand to other STEM subjects; integrate AR in low-resource settings; explore gender effects; teacher co-design	Expand to larger, diverse samples; apply to more complex astronomy; longitudinal research; teacher involvement	Test with larger groups; apply to other science topics; introduce control group; conduct long-term studies; involve teachers
Tools & AR Applications	Commercial AR apps (StarWalk2®, Star Tracker®, Solar System AR Core®, NASA's Spot the Station®, AR books)	Custom mobile AR apps, 3D models, AR markers	*SolarSystemRA* app (Unity + Vuforia, marker-based 3D planets)	Assemblr Edu app (AR barcodes in printed book)
Limitations	One school only; no long-term tracking; pre-existing AR tools, not custom; no qualitative feedback	Focus on one grade; short duration (8 weeks); technical issues; limited teacher role	Small sample (n=60); one NGO program; basic astronomy only; no long-term analysis	Small sample (n=46); one school; no control group; limited subject scope

THANK YOU!